

Technology #11

Gravity Compensation Hydraulic Damper for
Postural Transients

Engineering Technology-Transfer Dossier

CONFIDENTIAL

Technology Maturity: ENGINEERING_DEFINITION

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Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

Gravity Compensation Hydraulic Damper for Postural Transients

WHY DOES IT MATTER?

Postural ICP excursions cause over-drainage and under-drainage

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 5 governing equations, 9 design inputs, 7 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in Hydraulic Valve Mechanics + Proportional Damping. 6-step build plan with specific test articles and acceptance criteria. Target buyer: Neuroshunt manufacturer; valve engineering company.

WHAT DOES THE BUYER GET?

Proportional damper concept + mechanism description; Gravity head + damper + orifice governing equations; Bench test protocols (c_h measurement, postural simulation, aging, C_{pk}); External precedent catalog (ASD literature: PMC9133390, Miethke ShuntAssistant); Critical UNKNOWN disclosure (optimal c_h , comparative advantage over ASD); This engineering dossier with failure analysis + build plan

WHAT DOES THE BUYER HAVE TO BUILD?

Production damper element design (geometry, material, tolerances); Manufacturing process for damper + gravity reference + valve integration; Comparative clinical evidence vs existing ASDs; Regulatory submission (510(k) with substantial equivalence to ASD, or De Novo); Clinical validation (ICP excursion endpoint); Long-term reliability testing

WHAT IS THE NEXT DECISIVE EXPERIMENT?

Damper element prototypes (various geometries)

WHAT WOULD MAKE US KILL IT?

Damper coefficient cannot be tuned to both attenuate transients AND maintain minimum drainage floor

WHAT TRANSACTION COULD MAKE SENSE?

Sponsored Validation

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 5 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 6-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: ENGINEERING_DEFINITION

NEXT ENGINEERING STEP: Damper element prototypes (various geometries)

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - Damper element prototypes (various geometries) - 8 weeks

Basis: Build plan WP. Scope: c_h vs flow rate. Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - Damper element aging specimens - 12-24 weeks (chronic)

Basis: Build plan WP. Scope: c_h vs time. Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-05 - Biocompatibility specimens - 12 weeks (external)

Basis: Build plan WP. Scope: ISO 10993 series. Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 5 equations. Key assumptions: CSF is incompressible (valid at these pressures); Damper response is linear ($F = c \cdot v$); Gravity reference is stable; No cavitation; Catheter compliance negligible. UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

5 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: Damper element prototypes (various geometries). Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: Proportional hydraulic damper attenuates postural pressure transients continuously (vs ASD on/off switching)

POTENTIAL VALUE DRIVER: Addresses: Postural ICP excursions cause over-drainage and under-drainage

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: Neuroshunt manufacturer; valve engineering company

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: Hydraulic Valve Mechanics + Proportional Damping

Disciplines: Hydraulic valve engineering, Proportional control (mechanical), Fluid mechanics (gravity-driven flow), Polymer engineering (damper element)

Architecture: CSF shunt with hydraulic damper that proportionally attenuates postural pressure transients. Unlike fixed anti-siphon devices (ASD) that switch open/closed, this provides continuous proportional damping based on gravity head.

Subsystems:

SS-01: Damper element (proportional) - Hydraulic resistance that varies with flow rate or pressure differential; provides damping coefficient c

SS-02: Gravity reference chamber - References valve setting to gravity (postural); provides $dP_{gravity} = \rho * g * dh$

SS-03: Main catheter lumen - Carries CSF flow from ventricle to peritoneum

SS-04: Outlet (peritoneal) catheter - Drains CSF to peritoneal cavity

2. Mechanism Architecture

Physical Changes: Postural change creates hydrostatic pressure differential; damper element provides hydraulic resistance proportional to flow, attenuating the transient

Key Physics: Gravity head: $dP_{gravity} = \rho * g * dh$. Damper equation: $F = c * v$ (force proportional to velocity). For hydraulic damper: $dP_{damper} = c_h * Q$, where c_h is hydraulic damping coefficient.

3. Governing Engineering Model

Summary: Hydraulic damper provides flow-proportional resistance. Postural transients excite the system; damper coefficient c_h determines settling time and overshoot.

Equations:

$dP_{gravity} = \rho * g * dh$ [hydrostatic pressure differential]

$F = c * v$ [linear viscous damper]

$dP_{total} = dP_{valve} + dP_{gravity} - dP_{damper}$

$dP_{damper} = c_h * Q$ [hydraulic damping]

Settling time: $\tau = I_h / c_h$ [first-order response]

Assumptions:

- CSF is incompressible (valid at these pressures)
- Damper response is linear ($F = c*v$)
- Gravity reference is stable
- No cavitation
- Catheter compliance negligible

Boundary Conditions:

- Postural range: supine to upright (dh up to ~50 cm H₂O ~ 37 mmHg)
- ICP safety range: 5-20 mmHg normal
- Flow range: 0.05-0.5 mL/min typical

Failure Regimes:

- c_h too low — no damping benefit; behaves like fixed valve
- c_h too high — underdrainage; chronic intracranial hypertension
- Damper element failure — reverts to fixed valve behavior

- Gravity reference drift — postural compensation inaccurate

4. Design Inputs

ID	Input	Value	Origin
DI-001	Clinical need	Postural ICP excursions cause symptoms and complications; existing ASDs VERIFIED don't damp proportionally (R332)	VERIFIED
DI-002	Functional requirement	Proportional hydraulic damping of postural pressure transients	MODELLED
DI-003	Postural head range	EXTERNAL_PRECEDENT (0-50 cm H2O typical cranial-to-peritoneal)	EXTERNAL_PRECEDENT
DI-004	Flow regime	Laminar, Re << 2300 (damper element flow through proportional orifice)	COMPUTATIONALLY_SUPPORTED
DI-005	Damper coefficient target	UNKNOWN — to be optimized	UNKNOWN
DI-006	Response time target	MODELLED (< postural transient ~seconds)	MODELLED
DI-007	Biocompatibility (ISO 10993)	UNKNOWN	UNKNOWN
DI-008	Sterilization	UNKNOWN — likely EtO or gamma	UNKNOWN
DI-009	Comparative clinical endpoint vs ASD	UNKNOWN — needs clinical input	UNKNOWN

5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	Damper element design (geometry, material, c_h target)	ABSENT	c_h target; Material selection; Geometry
DO-002	Gravity reference chamber design	ABSENT	Reference mechanism; Stability spec
DO-003	Integrated valve + damper assembly	ABSENT	Component layout; Housing design
DO-004	Bench test protocol (postural simulation)	CONCEPTUAL	

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
Damper coefficient c_h	UNKNOWN	mmHg/(mL/min)	Design choice	Bench flow test
Gravity head range	EXTERNAL_PRECEDENT (postural H2O) to INTERNAL_H2O (resting physiology)	cm H2O	Robust design study	50 cm H2O
Response time tau	UNKNOWN — target < postural transient duration	seconds	MODEL_DERIVED	
Damper element material	UNKNOWN	n/a	Material selection	
Damper element geometry	UNKNOWN	mm	Design choice	
Drainage floor (minimum Q)	UNKNOWN — MODEL_DERIVED candidate	0.05 mL/min	Critical need	

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty
Damper element degradation	Mechanical fatigue or material changes	Material changes & lifetime design	Aging study with periodic checks	UNKNOWN degradation rate
Insufficient damping (c_h too low)	Damper provides negligible resistance	Increase c_h; redesign damper	Bench flow test	UNKNOWN optimal c_h
Excessive damping (c_h too high)	Damper hinders underdrainage	Reduce c_h; ensure drainage	Bench flow test	UNKNOWN threshold
Gravity reference drift	Reference chamber leaks or seals	Sealed reference; mechanical stops	Aging study	UNKNOWN achievable stability
Air entrapment	Air bubble in damper chamber	Priming process; sparging design	Air entrapment bench test	UNKNOWN clinical priming reliability
Catheter obstruction (Separate)	tissue blocks lumen	Standard obstruction mitigation	Obstruction bench	UNKNOWN whether damper affects obstruction rate
Manufacturing tolerance	Damper geometry tolerance	Cpk study; tight tolerance stack	Cpk measurement	UNKNOWN achievable Cpk

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
Damper element degradation	Polymer creep/fatigue changes c_h	Material selection + derating	UNKNOWN degradation rate
Insufficient damping	c_h too low	Increase c_h; redesign	UNKNOWN optimal c_h
Excessive damping	c_h too high; underdrainage	Reduce c_h; drainage floor	UNKNOWN threshold
Gravity reference drift	Chamber leak or shift	Sealed reference; mechanical stops	UNKNOWN achievable stability
Air entrapment	Air bubble compresses, spoils damping	Priming protocol; de-airing design	UNKNOWN clinical priming reliability
Catheter obstruction	Debris/tissue blocks lumen	Standard obstruction mitigation	UNKNOWN damper effect on obstruction rate
Manufacturing tolerance violation	Damper geometry tolerance > margin	Cpk study + tolerance analysis	UNKNOWN achievable Cpk

9. Verification Strategy

ID	Requirement	Method	Acceptance	Result
V-001	c_h within target range	Bench flow test	c_h in target	NOT_TESTED
V-002	ICP excursion attenuation >= 30% vs fixed valve bench	Reduced valve bench	>= 30% reduction	NOT_TESTED
V-003	Minimum drainage Q >= Q_min	Bench flow at worst-case	Q >= Q_min	NOT_TESTED
V-004	c_h stability over lifetime	Aging study	Drift < 20%	NOT_TESTED
V-005	ISO 10993 compliance	ISO 10993	Pass	NOT_TESTED

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VAL-001	ICP excursion reduction in patient cohort	Clinical trial (IDE)	Significant reduction vs fixed valve	NOT_PERFORMED
VAL-002	Clinical benefit	Clinical trial	Symptom or complication reduction	NOT_PERFORMED

11. Materials

Component	Candidate Material	Source	Verification Required	Status
Damper element	Silicone elastomer	Standard CSF shunt material; tuneable	ISO 10993 elastic characterization	CANDIDATE
Damper element	Polyurethane	Some shunt components use polyurethane	ISO 10993 + c_h + fatigue	CANDIDATE
Gravity reference chamber	Polymer housing + radiopaque marker	Standard ASD construction	ISO 10993 + radiopacity per ASTM F404	CANDIDATE

12. Bill of Materials

Item	Description	Qty	Type	Material	Supplier	Criticality	Verification
01	Damper element (proportional)	1	CUSTOM_COMPONENT	UNKNOWN — likely silicone	UNKNOWN	CRITICAL	c_h measurement + fatigue
02	Gravity reference chamber	1	CUSTOM_COMPONENT	UNKNOWN — likely polymer	UNKNOWN	CRITICAL	Stability test
03	Valve housing	1	CUSTOM_COMPONENT	UNKNOWN — likely silicone	UNKNOWN	MEDIUM	Dimensional + biocompatibility
04	Inlet + outlet catheters	2	CUSTOM_COMPONENT	UNKNOWN — silicone	Multiple	HIGH	Dimensional + biocompatibility

13. Manufacturing

Status: ENGINEERING_CANDIDATE — no proportional damper process validated

Candidate Processes:

- Injection molding (damper element + housing): Standard CSF shunt manufacturing - Damper geometry requires tight tolerance
- Assembly (damper + reference + valve + catheters): Standard shunt assembly - Custom fixture for damper alignment
- Radiopaque marker integration: Standard radiopacity process - Required for post-implant visualization

14. External Evidence

Source 1: Antisiphon device: A review of existing mechanisms and

URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC9133390>

Excerpt: by AP Huang · 2021 · Cited by 33 — The VP shunt connects the intracranial and intra-abdominal compartments, technically becoming a siphon when the system is in the vertical. Since the 1970s, ... Read more

Source 2: Persistent hydrocephalus due to postural activation of a

URL: <https://www.sciencedirect.com/science/article/abs/pii/S0967586816305094>

Excerpt: by CL Craven · 2017 · Cited by 3 — Antigravity shunt components implanted cranially in bedridden hydrocephalus patients will produce underdrainage due to head flexion induced anti - gravity device

Source 3: Effect of antisiphon devices on ventriculoperitoneal shunt

URL: <https://thejns.org/pediatrics/view/journals/j-neurosurg-pediatr/32/1/article-p50.xml>

Excerpt: by N Tachatos · 2023 · Cited by 4 — Gravitational shunt valves in hydrocephalus to challenge the sequelae of over-drainage.

Source 4: Gravitational Valves in Hydrocephalus Management

URL: <https://encyclopedia.pub/entry/6627>

Excerpt: The negative shunt -induced ICP was attributed to the gravity -induced ' siphoning ' effects, Siphoning induces shunt over-drainage and is the cause of many ...

Source 5: Understanding Shunt Systems

URL: <https://www.hydroassoc.org/shunt-systems>

Excerpt: Fixed pressure valves regulate the pressure within the brain (called intracranial pressure or ICP) ... These are called anti - siphon or anti- gravitational devices. Read more

15. Transfer Boundary

YOU RECEIVE:

- + Proportional damper concept + mechanism description
- + Gravity head + damper + orifice governing equations
- + Bench test protocols (c_h measurement, postural simulation, aging, Cpk)
- + External precedent catalog (ASD literature: PMC9133390, Miethke ShuntAssistant)
- + Critical UNKNOWN disclosure (optimal c_h, comparative advantage over ASD)
- + This engineering dossier with failure analysis + build plan

YOU MUST DEVELOP:

- > Production damper element design (geometry, material, tolerances)
- > Manufacturing process for damper + gravity reference + valve integration
- > Comparative clinical evidence vs existing ASDs
- > Regulatory submission (510(k) with substantial equivalence to ASD, or De Novo)
- > Clinical validation (ICP excursion endpoint)
- > Long-term reliability testing

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- Optimal damper coefficient c_h — UNKNOWN
- Achievable c_h given manufacturing constraints — UNKNOWN
- Comparative advantage over existing ASDs — UNKNOWN (critical for clinical adoption)
- Damper element long-term stability — UNKNOWN
- Regulatory pathway (likely 510(k) with substantial equivalence to existing ASD, or De Novo) — UNKNOWN
- Clinical benefit magnitude for proportional damper — ICP excursion amplitude reduction vs existing ASD — UNKNOWN until comparative trial

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	Damper element prototype	Types (low speed) orifice flow sensor, pressure transducer with stable response	ICP, excursions	c_h drift < 20% over 12 weeks	Stability report	8 weeks
WP-02	Integrated valve + damper	Integrable gravity reference	ICP, excursions	30% reduction in excursion amplitude	Integrated report	12 weeks
WP-03	Damper element aging	Specimen chamber at 37°C in vss frame	c_h in vss frame	c_h drift < 20% over 12 weeks	Stability report	12-24 weeks (chronic)
WP-04	Manufacturing process	Exposure/holding line	Critical dimensions, c_pk	c_pk >= 1.33 for critical dimensions	Manufacturing report	12 weeks
WP-05	Biocompatibility specification	ISO 10993 test lab	ISO 10993 series	Pass	Biocompatibility report	4 weeks (external)
WP-06	Air entrapment test article	Bench flow loop with air ejector	Effect of air on c_h	Damper tolerates specified air volume	Air entrapment report	4 weeks

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 5 governing equations and a 6-step build plan. The next experiment is clearly specified. Target buyer: Neuroshunt manufacturer; valve engineering company.

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: Damper coefficient cannot be tuned to both attenuate transients AND maintain minimum drainage floor

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: Damper coefficient cannot be tuned to both attenuate transients AND maintain minimum drainage floor

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.